Diego Gerardo Sánchez Moreno

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Summary

Robotics and embedded systems engineer focused on control, perception, and real-world deployments. Co-founder & Project Director at Autónoma (AutoNomaMX), with three POS systems in production. Full portfolio with projects and demos: diegogerardo.com.

Education

B.S. in Robotics and Digital Systems Engineering
Tecnológico de Monterrey, Campus Querétaro — 2022 – 2026
Professional license (Cédula Profesional) 15796458 — SEP, Mexico, 2026

Experience / Research

Autónoma (AutoNomaMX), Querétaro — Co-founder & Project Director
Jan 2026 – Present

- Shipped three offline-first POS systems in production (restaurant, coffee shop, optical shop) on a multi-tenant PostgreSQL/Supabase backend with Row-Level Security.
- Deployed CI/CD with self-hosted GitHub Actions runners on the production hardware (Raspberry Pi, Ubuntu) with automated smoke tests on every push.
- Integrated POS hardware: ESC/POS thermal printers (USB and network), Code128 barcode labels on Brother DK-1204, touch monitors, and UPS.
- Directed Phase 1 of an optical-shop deployment under PMBOK (charter, WBS, CPM, signed delivery, client training, closure with $0 cost deviation).

Hybrid Soft Robotics Lab, Tec de Monterrey (Querétaro) — Soft Robotics Control Platform
2024 – Present

- Designed a bi-stable pneumatic driver with closed-loop pressure/vacuum control and safety interlocks.
- Implemented micro-ROS on ESP32 (UART) with pressure sensing via I2C (ADS1115).
- Achieved near-zero steady-state error (0.000 kPa / -0.023 kPa) at 47 kPa and -30 kPa setpoints with PI control (Ts = 0.02 s).
- Built Python SDK + GUI for reproducible experiments and logging; analyzed control data with MATLAB.

Instituto Asunción de Querétaro — Software Developer / IT
Jun 2026 – Present

- Sole author of two web systems in production (SATAG, vehicle access; SEVAD, anonymous QR service evaluation) on PostgreSQL with Row-Level Security, plus a cryptographic prototype (SIVEA) using rotating Ed25519 credentials.
- Real-time data integration with a third-party access-control system and parking analytics from gate-reader logs.

Patio Cinco (Querétaro) — Production Platform
2025 – 2026

- Delivered a production system for memberships, billing, and customer operations used by real clients.

Selected Projects

PuzzleBot Autonomous Mobile Robot (ROS 2)
Jan–Jun 2025

- Integrated micro-ROS, encoder-based odometry, and YOLOv8 perception on Jetson Nano.
- Implemented telemetric distance tracking and validated autonomous motion.

Awards & Certifications

- Project Management program — Tecnológico de Monterrey (2026): Fundamentals (integrative exam), Scope, Cost, Time & Resources, Execution/Monitoring & Control
- Best Poster Award — Exploring Soft Robotics (Dec 2025)
- NVIDIA DLI: Fundamentals of Deep Learning (Jun 2025)
- OpenCV Bootcamp (Apr 2025)
- Google Cloud Computing Fundamentals (Mar 21, 2023)

Technical Skills

- **Languages:** C++, Python, TypeScript, VHDL, MATLAB
- **Robotics/Control: ROS 2 (Humble)**, **micro-ROS**, PID/PI, odometry, Gazebo
- **Perception:** OpenCV, YOLOv8
- **Hardware:** ESP32, Jetson Nano, Raspberry Pi, Arduino, FPGA
- **Production systems:** Next.js/React, FastAPI, Docker, Supabase/PostgreSQL (RLS), CI/CD (GitHub Actions self-hosted), offline-first sync

Languages

Spanish (Native), English (B2; technical documentation and technical conversations)